

MICHIGAN DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY

INTEROFFICE COMMUNICATION

TO: Tom Braum, Permits Section, Water Resources Division

FROM: Glen Schmitt, Permits Section, Water Resources Division

DATE: October 17, 2022

SUBJECT: Cadillac Wastewater Treatment Plant (WWTP)
Permit Review Biologist Recommendations
National Pollutant Discharge Elimination System (NPDES) Permit No. MI0020257

Pursuant to your request, we have reviewed the NPDES permit reissuance application for the Cadillac WWTP facility located in Cadillac, Michigan. The facility is requesting to continuously discharge a maximum of 3.2 million gallons per day (MGD) (this flow is not to be considered a limitation or actual capacity) of treated municipal wastewater from Monitoring Point 001A through Outfall 001. Outfall 001 discharges to the Clam River.

The Clam River has a lowest 95 percent exceedance, harmonic mean, and 90dQ10 low flows of 2.6, 13, and 3.4 cubic feet per second (cfs), respectively (Low Flow File No. 10194 [Wexford County]). The Clam River designated as warmwater within this reach and is not protected as a public water supply source.

Sources

Sources of information used for this review to develop the water quality-based effluent limits (WQBELs) include the facility's NPDES permit reissuance application, the current NPDES permit, the Permits Section Toxics facility file, discharge monitoring report data from November 2017 through August 2022, a Point Source Monitoring Survey conducted on May 9, 2018, and United States Geological Survey topographical maps. Based upon our review of this information, we have the following recommendations:

Recommendations for Outfall 001

- 1) The current NPDES permit requires daily flow reporting. The Outfall 001 final effluent flow from November 2017 through August 2022 ranged from 1.09 to 2.97 MGD, averaging 1.76 MGD as compared to the design flow of 3.2 MGD. Based on this information, we recommend the draft NPDES permit retain the daily monitoring requirement for the Outfall 001 final effluent flow.
- 2) The current NPDES permit includes daily minimum and daily maximum final effluent limitations for pH of 6.5 and 9.0 Standard Units (S.U.), respectively. Compliance with these limits is collected as five times weekly grab samples. Levels of pH in the Outfall 001 final effluent from November 2017 through August 2022 ranged from 6.2 to 8.4 S.U., averaging 7.2 S.U. There were two days during this period when the daily minimum limit of 6.5 S.U. was exceeded. Based on this information, we recommend the draft NPDES permit retain the current final effluent limits and monitoring requirements for pH.
- 3) The current NPDES permit includes a monthly average total phosphorus concentration limit of 0.5 milligram per liter (mg/l) and a loading limit of 13 pounds per day (lbs./day) from May through March and 0.65 mg/l (17 lbs./day) for April. Compliance with these limitations is collected five times weekly as 24-hour composite samples. Concentration

and loading of total phosphorus in the Outfall 001 final effluent from November 2017 through August 2022 ranged from 0.05 mg/l to 52.54 mg/l, averaging 0.42 mg/l as a concentration and 0.11 lbs./day to 33.7 lbs./day, averaging 2.9 lbs./day as a loading. The monthly average concentration limit was exceeded in May 2022. Based on this information, we recommend the draft NPDES permit retain the final effluent limitations and monitoring requirements for total phosphorus.

- 4) It is understood that ultraviolet disinfection is used to disinfect the final effluent. There is no information indicating chlorine is used for disinfection. Based on this information, we have no recommendations for total residual chlorine. However, if chlorine is used, we recommend the draft NPDES permit include a daily maximum WQBEL of 38 micrograms per liter (ug/l) with daily compliance monitoring and samples collected as grab samples.
- 5) The current NPDES permit includes a monthly average and daily maximum WQBEL for total copper of 11 micrograms per liter (ug/l) (0.3 lbs./day) and 31 ug/l (0.82 lbs./day), respectively. Compliance with these limitations is conducted as weekly 24-hour composite samples. Concentrations of total copper in the Outfall 001 final effluent from November 2017 through August 2022 ranged from not detected (quantification level ranged from 0.075 ug/l to 10 ug/l) to 24 ug/l. Total copper data collected from November 2017 through August 2019 was not used as the quantification level report (<50 ug/l) is not considered sufficiently sensitive. These data indicate there is not a reasonable potential for total copper to be discharged from the Outfall 001 final effluent at concentrations exceeding Michigan Water Quality Standards (WQS). We recommend the draft NPDES permit remove the limitations for total copper. However, due to concerns with elevated concentrations of total copper, we recommend the draft NPDES permit include a weekly monitoring requirement with samples collected as 24-hour composite samples and analyzed using an approved USEPA analytical method capable of achieving a quantification level of 1.0 ug/l. Due to the quantification levels reported during the period reviewed, we do not recommend monitoring frequency reduction language to be included in the draft NPDES permit.
- 6) The current NPDES permit includes quarterly monitoring requirement for total nickel with samples collected as 24-hour composite samples. Concentrations of total nickel in the Outfall 001 final effluent from November 2017 through August 2022 ranged from not detected (quantification level ranged from 0.075 ug/l to 5.0 ug/l) to 7.8 ug/l. These data indicate there is not a reasonable potential for total nickel to be discharged from the Outfall 001 final effluent at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit include monitoring for total nickel in the Additional Monitoring Requirements.
- 7) The current NPDES permit includes a monthly average WQBEL for total selenium of 5.7 ug/l (0.15 lbs./day). Compliance with this limitation is conducted as monthly 24-hour composite samples. Concentrations of total selenium in the Outfall 001 final effluent from November 2017 through August 2022 ranged from not detected (quantification level ranged from 0.081 ug/l to 5.0 ug/l) to 2.0 ug/l. These data indicate there is not a reasonable potential for total selenium to be discharged from the Outfall 001 final effluent at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit remove the final effluent limitations for total selenium and include monitoring for total selenium as part of the Additional Monitoring Requirements.
- 8) The current NPDES permit includes a monthly average WQBEL for Bis(2-ethylhexyl) phthalate of 31 ug/l (0.82 lbs./day). Compliance with this limitation is conducted as monthly 24-hour composite samples. Concentrations of Bis(2-ethylhexyl) phthalate in the Outfall 001 final effluent from November 2017 through August 2022 ranged from not detected (quantification level ranged from 3.9 ug/l to 4.7 ug/l) to 4.1 ug/l. These data indicate there is not a reasonable potential for Bis(2-ethylhexyl) phthalate to be discharged from the Outfall 001 final effluent at concentrations exceeding Michigan

WQS. We recommend the draft NPDES permit remove the final effluent limitations for Bis(2-ethylhexyl) phthalate and include monitoring for Bis(2-ethylhexyl) phthalate as part of the Additional Monitoring Requirements.

- 9) The current NPDES permit includes a Level Currently Achievable (LCA) Total Mercury limitation of 2.0 nanograms per liter (ng/l) and a loading limitation of 0.000053 lbs./day calculated as a 12-month rolling average. Compliance with this limitation is collected as quarterly grab samples. Part I.A.4 of the current NPDES permit includes requirements for a Pollutant Minimization Program for Total Mercury. Concentrations of total mercury in the Outfall 001 final effluent from November 2017 through August 2022 ranged from not detected (quantification level <0.5 ng/l) to 2.82 ng/l. Based on the final effluent data for total mercury, there is a reasonable potential for total mercury to be discharged from Outfall 001 at levels exceeding Michigan WQS. We recommend the draft NPDES permit retain the LCA of 2.0 ng/l (0.000053 lbs./day). Compliance with this limitation shall be collected as quarterly grab samples using USEPA method 1669 and analyzed using USEPA method 1631 with a quantification level of 0.5 ng/l. In addition, we recommend the PMP language in Part I.A.4 be retained in the draft permit. We do not recommend the draft NPDES permit include monitoring frequency reduction language for total mercury.
- 10) The current NPDES permit requires Whole Effluent Toxicity (WET) testing for acute and chronic toxicity as part of the conditions in Part I.A.3, Additional Monitoring Requirements with testing conducted using Fathead minnow and *Ceriodaphnia dubia* (*C. dubia*) as the test species for a total of eight tests (four tests on each species). Acute toxicity test results from November 2017 through August 2022 for *C. dubia* and Fathead minnow never measured any acute toxicity (all results were 0.0 acute toxic units (TUa)). Chronic toxicity for *C. dubia* ranged from 0.0 chronic toxic units (TUC) to 5.6 TUC with toxicity occurring during the two tests conducted in August 2019. No other tests demonstrated chronic toxicity. Only one of five tests were chronically toxic to Fathead minnow with a result of 1.1 TUC.

Based on this information, there is a reasonable potential for the Outfall 001 final effluent to be chronically toxic to aquatic life. We recommend the draft NPDES permit include a chronic and acute WET limitation of 1.1 TUC and 1.0 TUa, respectively. Compliance with these limitations shall be quarterly with samples collected as 24-hour composite samples and testing using Fathead minnow and *C. dubia* as the test species. Testing shall be in accordance with chronic toxicity testing methods. Appropriate language shall also be included in the draft NPDES permit to allow the permittee to request a monitoring frequency reduction to no less than annually after two years of quarterly testing has been completed based on the effective date of the NPDES permit.

- 11) Arsenic monitoring was conducted as part of the conditions in the current NPDES permit under Part I.A.3 Additional Monitoring Requirements. Concentrations of arsenic in the Outfall 001 final effluent from November 2017 through August 2022 ranged from not detected (quantification level of <1.0 ug/l) to 76 ug/l. These data indicate there is a reasonable potential for arsenic to be discharged at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit include a monthly average WQBEL of 16 ug/l (0.42 lbs./day). Compliance with this limitation shall be weekly with samples collected as 24-hour composite samples and analyzed using an approved USEPA analytical method capable of achieving a quantification level of 1.0 ug/l. Finally, we recommend the draft NPDES permit include monitoring frequency reduction language that allows the permittee to reduce the monitoring frequency to no less than monthly after two years of weekly monitoring based on the effective date of the NPDES permit.
- 12) Beryllium monitoring was conducted as part of the conditions in the current NPDES permit under Part I.A.3 Additional Monitoring Requirements. Concentrations of beryllium in the Outfall 001 final effluent from November 2017 through August 2022 ranged from

not detected (quantification level ranged from 0.4 ug/l to <1.0 ug/l) to 8.1 ug/l. These data indicate there is a reasonable potential for beryllium to be discharged at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit include a monthly average WQBEL of 4.1 ug/l (0.11 lbs./day). Compliance with this limitation shall be weekly with samples collected as 24-hour composite samples and analyzed using an approved USEPA analytical method capable of achieving a quantification level of 1.0 ug/l. Finally, we recommend the draft NPDES permit include monitoring frequency reduction language that allows the permittee to reduce the monitoring frequency to no less than monthly after two years of weekly monitoring based on the effective date of the NPDES permit.

- 13) Cadmium monitoring was conducted as part of the conditions in the current NPDES permit under Part I.A.3 Additional Monitoring Requirements. Concentrations of cadmium in the Outfall 001 final effluent from November 2017 through August 2022 ranged from not detected (quantification level <0.2 ug/l) to 9.5 ug/l. These data indicate there is a reasonable potential for cadmium to be discharged at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit include a monthly average WQBEL and daily maximum WQBEL of 4.1 ug/l (0.11 lbs./day) and 12 ug/l (0.32 lbs./day), respectively. Compliance with these limitations shall be weekly with samples collected as 24-hour composite samples and analyzed using an approved USEPA analytical method capable of achieving a quantification level of 0.2 ug/l. Finally, we recommend the draft NPDES permit include monitoring frequency reduction language that allows the permittee to reduce the monitoring frequency to no less than monthly after two years of weekly monitoring based on the effective date of the NPDES permit.
- 14) Silver monitoring was conducted as part of the conditions in the current NPDES permit under Part I.A.3 Additional Monitoring Requirements. Concentrations of silver in the Outfall 001 final effluent from November 2017 through August 2022 ranged from not detected (quantification level ranged from 0.2 ug/l to <0.5 ug/l) to 1.8 ug/l. These data indicate there is a reasonable potential for silver to be discharged at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit include a monthly average WQBEL of 1.4 ug/l (0.036 lbs./day). Compliance with this limitation shall be weekly with samples collected as 24-hour composite samples and analyzed using an approved USEPA analytical method capable of achieving a quantification level of 0.5 ug/l. Finally, we recommend the draft NPDES permit include monitoring frequency reduction language that allows the permittee to reduce the monitoring frequency to no less than monthly after two years of weekly monitoring based on the effective date of the NPDES permit.
- 15) Available cyanide monitoring was conducted as part of the conditions in the current NPDES permit under Part I.A.3 Additional Monitoring Requirements. Concentrations of available cyanide in the Outfall 001 final effluent from November 2017 through August 2022 ranged from not detected (quantification level ranged from 1.0 ug/l to 2.0 ug/l) to 8.4 ug/l. These data indicate there is a reasonable potential for available cyanide to be discharged at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit include a monthly average WQBEL of 5.9 ug/l (0.16 lbs./day). Compliance with this limitation shall be monthly with samples collected as a grab sample and analyzed using Method OIA-1677 with a quantification level of 2.0 ug/l. Finally, we recommend the draft NPDES permit include monitoring frequency reduction language that allows the permittee to reduce the monitoring frequency to no less than annually after two years of monthly monitoring based on the effective date of the NPDES permit.
- 16) Benzidine monitoring was conducted as part of the conditions in the current NPDES permit under Part I.A.3 Additional Monitoring Requirements. Concentrations of benzidine in the Outfall 001 final effluent from November 2017 through August 2022 ranged from not detected (quantification level of <0.1 ug/l) to 1.8 ug/l. These data indicate there is a reasonable potential for benzidine to be discharged at concentrations

exceeding Michigan WQS. We recommend the draft NPDES permit include a monthly average WQBEL of 0.12 ug/l (0.003 lbs./day). Compliance with this limitation shall be weekly with samples collected as a grab sample and analyzed using an approved USEPA analytical method capable of achieving a quantification level of 0.1 ug/l. Finally, we recommend the draft NPDES permit include monitoring frequency reduction language that allows the permittee to reduce the monitoring frequency to no less than annually after two years of weekly monitoring have been completed based on the effective date of the NPDES permit.

- 17) Bromomethane monitoring was conducted as part of the conditions in the current NPDES permit under Part I.A.3 Additional Monitoring Requirements. Concentrations of bromomethane in the Outfall 001 final effluent from November 2017 through August 2022 were all reported as not detected with quantification levels of <1.0 ug/l for two results and <5.0 ug/l for the other two results. The samples that reported a quantification level of <5.0 ug/l were reported with a quantification level that is not considered sufficiently sensitive when compared to the theoretical WQBELs. Based on this information, we recommend the draft NPDES permit include a quarterly monitoring requirement for bromomethane. Samples for bromomethane shall be collected as grab samples and analyzed using an approved USEPA analytical method capable of achieving a quantification level of 1.0 ug/l. Finally, we recommend the draft NPDES permit include monitoring frequency reduction language that allows the permittee to reduce the monitoring frequency to no less than annually after two years of quarterly monitoring have been completed based on the effective date of the NPDES permit.
- 18) The current NPDES permit includes 2X annual monitoring for Perfluorooctane sulfonate (PFOS) with samples collected as grab samples. Concentrations of PFOS in the Outfall 001 final effluent from November 2017 through August 2022 ranged from not detected (quantification level ranged from <1.6 ng/l to <1.9 ng/l) to 7.8 ng/l. These data indicate there is a reasonable potential for PFOS to be discharged from Outfall 001 at concentrations exceeding Michigan WQS. We recommend a monthly average WQBEL of 0.012 ug/l (0.00032 lbs./day). Compliance with this limitation shall be quarterly with samples collected as grab samples. Analysis for PFOS shall be with method ASTM D7979 or an isotope dilution method (sometimes referred to as Method 537 modified) at a quantification level of 2.0 ng/l. Language shall also be included to allow for a monitoring frequency reduction request to no less than annually to be submitted after 10 equally spaced samples over a minimum of three months have been collected.
- 19) The current NPDES permit includes 2X annual monitoring for Perfluorooctanoic acid (PFOA) with samples collected as grab samples. Concentrations of PFOA in the Outfall 001 final effluent from November 2017 through August 2022 ranged from 1.9 ng/l to 20 ng/l. These data indicate there is not a reasonable potential for PFOA to be discharged from Outfall 001 at concentrations exceeding Michigan WQS. We have no recommendations for PFOA.
 - a. The theoretical WQBELs for PFOA are 0.28 ug/l (0.0075 lbs./day) as a Monthly Average WQBEL and 15,000 ug/l (400 lbs./day) as a Daily Maximum WQBEL.
- 20) The current NPDES permit does not include any final effluent limitations or monitoring requirements for Perfluorobutanesulfonate (PFBS). Final effluent data collected from the Outfall 001 final effluent from November 2017 through August 2022 ranged from 3.1 ng/l to 58 ng/l. These data indicate there is not a reasonable potential for PFBS to be discharged from Outfall 001 at concentrations exceeding Michigan WQS. We have no recommendations for PFBS.
 - a. The theoretical WQBELs for PFBS are 1,100 ug/l (30 lbs./day) as a Monthly Average WQBEL and 240,000 ug/l (6,400 lbs./day) as a Daily Maximum WQBEL.
- 21) As part of the Chloride and Sulfate Implementation Plan that became effective in February 2021, we recommend the draft NPDES permit include a monthly monitoring

requirement for Chloride and Sulfate for the full term of the permit. Samples for Chloride and Sulfate shall be collected as 24-hour composite samples and analyzed using an approved USEPA analytical method capable of achieving a quantification level of 1,000 ug/l for Chloride and 2,000 ug/l for Sulfate. We do not recommend the draft NPDES permit include a monitoring frequency reduction as a sufficient dataset will be required for review at the next reissuance.

Other Recommendations

- 22) The recommendations contained herein are based on the Part 4 Water Quality Standards and Part 8 Water Quality-Based Effluent Limit Development for Toxic Substances. Treatment methods or economic concerns have not been addressed. We understand that other considerations may be used in deciding permit limits and requirements.